

Planning Report for MSc Project

**Multimodal control of a wearable dual-arm robotic system for
assisted object manipulation**

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1. Project Specification

Wearable supernumerary robotic limbs (SRLs) augment human capabilities and assist individuals with limited mobility by providing extra manipulation arms that move synchronously with the user [1, 4]. The MUlti-limb Virtual Environment (MUVE), known as the Dr. Octopus platform at Imperial College London, mounts up to dual lightweight Kinova 7-DoF robotic arms on a backpack. This enables coordinated object manipulation in daily environments while supporting multisensory integration [2].

The overall goal of the project is to enable safe and efficient control of up to dual supernumerary robotic arms using the fingers of one or two hands moving on a small physical mannequin. The mannequin acts as the master device, directly commanding the dual full-scale Kinova arms (the slaves) mounted on the Dr. Octopus backpack. The project is structured in three sequential stages: Stage 1 consists of physical prototyping with a figurine and real springs to determine optimal spring placement for intuitive force feedback together with simulation setup of the Kinova arm dynamics in Isaac Lab to validate virtual spring behaviour; Stage 2 comprises the complete simulation in Isaac Lab along with design of the physical mannequin master hardware with real miniature arms and the design of the master-to-slave control system for the dual Kinova arms; Stage 3 involves implementation of representative tasks like pick and place and their demonstration. This project specifically focuses on the master mannequin design and control mapping for the master-slave manipulator. Using the physical mannequin as the master device makes the overall teleoperation system intrinsically safe through direct finger interaction; and finally, the multimodal vision-based object detection and gesture-based intent interfaces completed previously by the team will be integrated where relevant.

The Kinova Gen3 arms support admittance and impedance control modes via the Kortex API, enabling virtual spring-damper dynamics that emulate the compliance of physical Series Elastic Actuators (SEAs) [3, 10]. A Series Elastic Actuator (SEA) places an elastic element, such as a spring, in series between a motor/gearbox and the output load. This design enables precise force control, increased shock tolerance, and safe, compliant interaction with humans, making them ideal for exoskeletons, prosthetics, and walking robots [9, 13].

The project delivers a closed-loop haptic control demonstration for coordinated object manipulation using the dual Kinova arms, with virtual springs providing intuitive force feedback.

Aims: To design, prototype, and implement a master-slave teleoperation system using a physical figurine mannequin with miniature arms that is controlled by the fingers of one or two hands, thereby enabling safe and efficient intuitive control of up to dual Kinova supernumerary robotic limbs on the Dr. Octopus platform.

Objectives: The objectives of this project are to conduct physical prototyping with a figurine and real springs to determine optimal spring placement for intuitive force feedback and to perform simulation of the Kinova arm dynamics in Isaac Lab or MuJoCo to validate virtual spring behaviour, to design the physical mannequin master hardware with real miniature motorised arms and to design the control mapping that commands the dual Kinova 7-DoF slave arms while ensuring stable master-slave synchronisation and haptic transparency, and to implement representative tasks such as coordinated pick-and-place operations and synchronised multi-arm motions and demonstrate them on the real Dr. Octopus platform after completion of the master hardware and control design. Both the master mannequin design and the slave control mapping will be developed in parallel to accelerate integration. This work advances SRL teleoperation by bridging physical miniature prototyping with haptic springs to full-scale wearable hardware on the Dr. Octopus platform [1, 6].

2. Ethical Analysis

This project is laboratory-based and involves only robotic hardware development and testing. No human subjects beyond voluntary, supervised researcher demonstrations (with informed consent) are required. No biological specimens or personal data are collected. The only human involvement consists of voluntary, supervised researcher demonstrations and a small user study of 8–10 laboratory volunteers, all of whom will provide written informed consent prior to participation and retain the right to withdraw at any time without consequence. Testing complies with Imperial College robotic safety protocols.

Risks are confined to minor mechanical hazards during arm motion, mitigated by Kinova safety features (joint limits, emergency stops, impedance-based compliance) and supervised operation. Long-term benefits include safer, more intuitive assistive robotics for mobility-impaired users and industrial augmentation, with negligible environmental impact through reuse of existing Kinova hardware. The work follows responsible robotics principles emphasising safety and human-centred design.

At the hardware level, the physical mannequin serves a dual purpose beyond control by interacting with the miniature figurine during development, the operator naturally becomes aware of geometrically possible contacts between the four full-scale Kinova arms and the human body. This embodied awareness informs safer arm trajectory design before any live testing occurs, effectively making the prototyping process itself a safety validation tool. All mechanical components, including 3D-printed parts, coil springs, and servo motors, are selected and assembled to eliminate sharp edges, pinch points, and failure modes that could cause injury during handling.

At the electronics level, all wiring, connectors, and power supplies operate within low-voltage safety thresholds. Current-limiting protections are implemented in hardware to prevent motor overcurrent during unexpected joint collisions, and all electronic enclosures are secured to prevent accidental contact during operation.

At the software level, the Kinova Kortex API is operated exclusively within manufacturer-specified joint torque and velocity limits. Impedance control parameters are first validated in simulation before deployment on physical hardware, ensuring no unsafe gain combinations reach the real system. Redundant emergency stop functionality is implemented in software alongside the physical E-stop buttons, with automatic velocity zeroing triggered upon communication loss between master and slave.

From a human-centred design perspective, the mannequin interaction paradigm inherently keeps the operator physically engaged with the system's geometry throughout development, maintaining continuous awareness of reachable body regions and promoting intuitive safe operation. The work therefore follows responsible robotics principles at every stage, embedding safety not as an afterthought but as a structural property of the design process itself.

The environmental footprint of this project is deliberately minimised. All Kinova Gen3 hardware is reused from existing laboratory inventory, avoiding the carbon and material cost of new manufacturing. Miniature arm components are fabricated using desktop 3D printing, with recyclable filament materials used wherever available. Most of the Electronic components will be sourced from existing laboratory stocks.

3. Literature Review

Supernumerary robotic limbs (SRLs) have emerged as a key technology for human augmentation, drawing interest from both assistive technology and industrial domains. Comprehensive reviews quantify their design parameters, actuation mechanisms, intent recognition strategies, and multimodal feedback approaches across a range of applications [1, 4]. The Dr. Octopus platform at Imperial College London exemplifies a wearable multi-arm system capable of supporting physical interaction in both virtual and real environments, mounting up to dual Kinova 7-DoF arms on a backpack frame [2]. Early SRL prototypes focused primarily on passive augmentation for specific tasks such as overhead assembly; subsequent systems incorporated active impedance control to improve safety and naturalness of interaction [4, 8].

Master-slave teleoperation has been widely adopted in redundant manipulators and SRLs to provide intuitive, low-latency control [7]. Bilateral teleoperation schemes, in which force and position information flow in both directions between master and slave, are considered essential for safe physical interaction and transparent haptic feedback [10]. Recent work on Kinova Gen3 teleoperation demonstrates effective mapping from human gestures or physical master devices to 7-DoF slave arms, highlighting the importance of current-based impedance control for interacting with mobile manipulators [10]. Physical prototyping with miniature models is a well-established approach to validate haptic feedback and control mapping before scaling to full hardware [12]. Guggenheim and Asada demonstrated that inherent haptic feedback from SRL limbs can meaningfully reduce cognitive load during teleoperation, motivating the use of spring-based compliance at the master stage [12]. Embodiment studies in virtual reality further show that physical congruence between the master device and the robot morphology enhances the user’s sense of agency over supernumerary limbs [8].

Haptic feedback is essential for closed-loop teleoperation and embodiment in SRLs. Series Elastic Actuators (SEAs) place a compliant element in series between the actuator and the load, enabling precise force control, shock tolerance, and safe human interaction; their principles directly inform the virtual spring design used in this project [9, 13]. Physical springs on the master device, as in the figurine prototyping stage, provide direct mechanical force reflection, while virtual springs implemented through impedance control in the Kinova slave arms replicate this compliance at scale [3, 9]. Pinardi et al. demonstrated that supplementary sensory feedback significantly improves the control quality and embodiment of SRLs during manipulation tasks, underscoring the value of the haptic channel in this system [14]. Simulation tools such as Isaac Lab enable safe offline tuning of impedance parameters and master-slave mapping before deployment on real hardware, reducing the risk of hardware damage during early integration [11]. Spatial considerations for near-body SRL interaction including collision avoidance and ergonomic placement have also been studied to ensure the dual-arm configuration remains practical during daily tasks [15].

Approach	Control Mechanism	Strengths	Limitations	Reference
Direct master-slave mapping	Scaled miniature master device physically commands full-scale slave joints via position mapping	Highly intuitive; preserves natural hand motion; low end-to-end latency (~120 ms); no cognitive abstraction layer required	Geometric scaling must be precisely calibrated; kinematic dissimilarity between master and slave introduces mapping errors	[12]

Gesture and VR-based teleoperation	Vision or inertial sensors capture hand/body gestures and map to slave joint commands	Completely hands-free; no physical master device required; adaptable to different robot morphologies	Higher cognitive load due to absence of haptic feedback; prone to drift and occlusion; poor transparency during contact tasks	[5]
Virtual springs via impedance control	Kinova Kortex API impedance mode emulates spring-damper dynamics at slave joint level	Hardware-native; no additional actuators needed; stiffness and damping tunable in software; compatible with existing Kinova SDK	Requires careful gain tuning to avoid instability; virtual compliance cannot fully replicate physical spring feel; latency-sensitive	[3]
Physical springs on miniature master	Coil springs at master joint locations provide direct mechanical force reflection to operator fingertips	Passive, inherently safe, and zero-latency haptic feedback; no electronics required at master stage; directly encodes postural strain	Miniature spring fabrication and calibration is labour-intensive; spring stiffness fixed at manufacture; limited to reflecting gross postural state rather than fine contact forces	[9]
Simulation-in-the-loop tuning	Master-slave mapping and impedance parameters validated in Isaac Lab before hardware deployment	Safe offline parameter optimisation; reduces hardware damage risk; enables rapid iteration of control gains	Sim-to-real gap means tuned parameters require further adjustment on physical hardware; does not capture all real-world dynamics	[11]
EMG-based intent recognition	Surface electromyography signals from the operator's muscles decoded to infer intended arm motion	Intuitive biological control signal; hands remain free for primary tasks; low learning curve for trained users	Susceptible to electrode placement variability, muscle fatigue, and skin impedance changes; requires	[4]

			user-specific calibration	
			Requires reliable object detection pipeline;	
Shared autonomy with vision integration	Robot autonomously assists with low-level positioning while operator retains high-level directional intent	Reduces operator workload during repetitive or precise sub-tasks; improves task success rate	autonomous layer may conflict with operator intent during unexpected scenarios	[5]
			Added compliance reduces positioning bandwidth; tuning stiffness-	
Spring-based SEA compliance at slave	Series elastic actuator principles applied at Kinova joint level via software-defined virtual stiffness	Provides shock tolerance and passive safety; enables precise torque control; well-studied theoretical framework	bandwidth trade-off is non-trivial for multi-arm coordination	[9]

Table 1: Control strategies in supernumerary and teleoperated robotic systems. Master-slave with miniature models and hybrid physical/virtual springs aligns with SEA principles for the Dr. Octopus platform [9, 13]

Literature reveals a gap: few studies combine a custom miniature dual-arm master mannequin (with physical springs) and direct teleoperation of a wearable dual-arm SRL platform for coordinated manipulation tasks [1, 6]. Existing multi-arm SRL studies either rely on gestural or VR-based interfaces with higher cognitive load, or use single-arm master-slave configurations that do not scale to dual-limb coordination [4, 7]. Moreover, prior work rarely integrates SEA-style physical compliance at the miniature master stage with virtual spring impedance at the full-scale slave stage simultaneously; this dual-layer compliance strategy is a distinguishing feature of the present project [9, 12]. This project addresses the identified gap through a three-stage process of physical prototyping, parallel master-and-control design, and full-system task demonstration on the Dr. Octopus platform.

4. Implementation Plan

Work completed to date has focused on Stage 1 of the project. Physical prototyping has been carried out by placing real springs on a figurine model to determine optimal attachment points for intuitive force feedback, and simulation and design setup of the Kinova 7-DoF arm dynamics has begun in Isaac Lab to validate virtual spring behaviour.

The remaining work will proceed with Stage 2 and Stage 3. In Stage 2, which will run in parallel, Simulation of the dual arms in Isaac Lab the physical mannequin master hardware with real miniature motorised arms will be designed and constructed while the master-to-slave control mapping for the dual Kinova slave arms is simultaneously developed and simulated. In Stage 3, following completion of the master hardware and control design, representative tasks will be implemented on the real Dr. Octopus platform and demonstrated; these tasks include coordinated pick-and-place operations and synchronised multi-arm motions such as dancing sequences. Full system integration, safety validation, laboratory testing, performance tuning, documentation, and preparation of the final demonstration video will conclude the project. Key milestones include validation of virtual spring parameters and completion of the basic master mannequin and control mapping by the end of May 2026, operational master-slave synchronisation on two arms by the end of June, full dual-arm teleoperation and task demonstrations by the end of July, and complete evaluation by the end of August. Fallback positions have been identified: if miniature arm fabrication delays occur, a simplified 2-DoF prototype per arm will be used for initial mapping tests; if real-time slave synchronisation proves unstable, conservative scaling factors validated on the physical figurine will be applied before full deployment.

5. Risk register

Risk ID	Risk Description	Likelihood	Impact	Mitigation Strategy
R1	Mismatch between miniature master kinematics and full-scale slave	Medium	High	Direct geometric scaling from figurine tests; use simulation for validation
R2	Latency or instability in master-slave mapping	Low	High	Conservative gains and Isaac Lab/MuJoCo tuning; Kinova safety limits
R3	Hardware access delays for Dr. Octopus	Medium	Medium	Parallel simulation work; fallback to single-arm testing
R4	Integration with team's vision/gesture modules	Low	High	Modular API; weekly team sync
R5	Safety during dual-arm teleoperation demo	Low	High	Redundant E-stops; supervised testing only
R6	Miniature arm motor or spring fabrication delays	Medium	Medium	Use off-the-shelf servo motors and 3D-printed parts; early prototyping
R7	Poor haptic transparency (force reflection mismatch)	Medium	High	Calibrate physical springs on figurine and transfer parameters to virtual springs
R8	Real-time communication lag between master and slave	Low	High	Use high-speed Ethernet/ROS2; fallback to wired direct connection

R9	Overloading Kinova arms during multi-arm coordination	Low	High	Limit payload and velocity in control software; monitor torque feedback
R10	User finger fatigue or discomfort during long demos	Medium	Medium	Ergonomic mannequin handle design; short test sessions with breaks

Table 2. Identified main risks associated with achieving the objectives and deliverables, labelled them in terms of likelihood and impact, and detail your mitigation strategy.

6. Evaluation

Success of the project will be measured through a combination of quantitative performance metrics and qualitative user feedback. Master-slave tracking accuracy will be assessed by comparing miniature master joint angles to slave end-effector positions, targeting positional errors below 10 cm and angular errors below 15 degrees. Haptic transparency will be verified by ensuring force reflection errors remain below 0.8 N via the physical springs on the mannequin and Kinova torque feedback. Task-level metrics will include coordinated pick-and-place success rate above 80 percent and average completion time under 45 seconds per object, as well as synchronised motion accuracy for dancing demonstrations (measured by phase lag below 200 ms across arms). Closed-loop stability will be confirmed by oscillations below 8 percent overshoot and end-to-end latency under 200 ms. In addition, a small user study involving 8-10 laboratory volunteers will gather qualitative impressions of haptic intuitiveness and embodiment when controlling the dual arms via finger movements on the mannequin. Experiments will use standardised household objects in a controlled lab arena, comparing baseline position control against the full master-slave system with haptic feedback. Simple statistical comparisons (e.g., average and standard deviation) will be used to evaluate differences between conditions.

7. Preliminary Results

Stage 1 (physical figurine with real springs and Kinova arm primary simulation) is complete: optimal spring placement points were identified for balanced compliance, and initial virtual spring models in Isaac Lab are setup to show stable impedance behaviour for light payloads. Basic Design of CAD Model for Figurine with Frame is setup for slave position control. Simulation and Design of the physical mannequin master and control mapping (Stage 2) is now underway. Early tests with physical; dummy model indicate promising force reflection for intuitive finger-based multi-arm coordination.

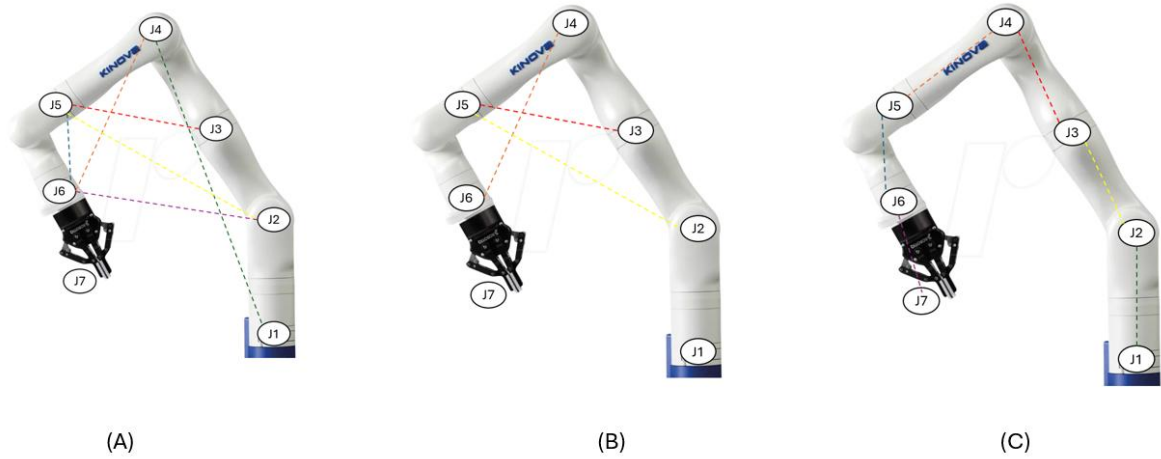


Figure 1: Spring attachment configurations on the Kinova Gen3 7-DoF arm figurine (J1–J7). **(A)** Full connectivity: all geometrically feasible inter-joint spring connections, providing maximum force information at the cost of coupled, ambiguous signals. **(B)** Informative subset: connections retained based on single-joint deflection dominance, eliminating cross-coupled springs that respond to multiple degrees of freedom simultaneously. **(C)** Optimal configuration: the minimal spring set that satisfies both postural observability and signal separability, enabling the operator to infer arm configuration comfort and joint proximity through clean, independently interpretable haptic feedback at the master fingertips.

To determine optimal spring placement on the miniature figurine master, three configurations of inter-joint spring connections were systematically evaluated across the seven joints (J1–J7) of the Kinova Gen3 arm morphology, as shown in Figure 1. For seven joints, the maximum number of 21 unique pairwise connections.

Configuration A retains a representative subset of these, connecting non-adjacent joints across the kinematic chain. While this maximises force information richness, it introduces mechanical coupling ambiguity the contribution of individual joint displacements to spring deformation cannot be cleanly isolated, making signal attribution unreliable.

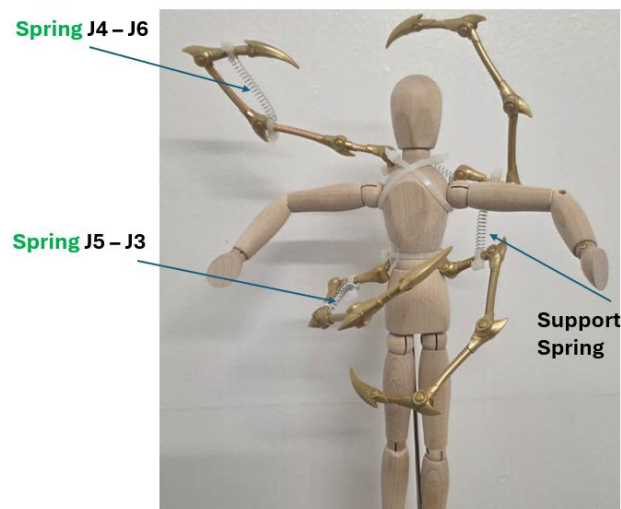


Figure 1: Physical Figurine Model with Springs Configuration B

Configuration B retains only connections that produce mechanically meaningful and attributable spring deflection signals. Specifically, a connection is kept only if its deformation is dominated by the motion of a single joint, rather than being spread across multiple joints simultaneously. This eliminates cross-coupled connections whose force signals would be ambiguous at the operator's fingertips, leaving a reduced set of springs each acting as a clean proprioceptive channel for one dominant degree of freedom. In the Figure 2. The figurine also has a support spring which separates the upper arms with lower arms.

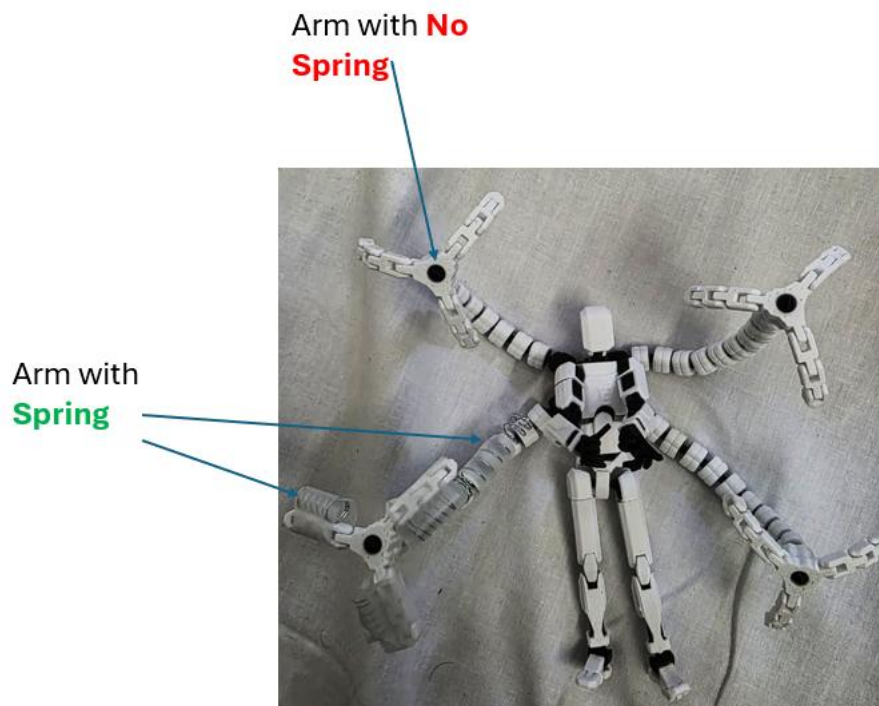


Figure 3: Physical Figurine Model with Springs Configuration C

Configuration C represents the optimal minimal set identified through this process. It provides sufficient coverage to infer the global postural state of the arm including whether a given configuration is ergonomically comfortable or approaching a joint limit while keeping the haptic signals at the master fingertips clean and separable. Each spring in Configuration C responds predominantly to one degree of freedom: the J5–J6 connection encodes elbow flexion state, J3–J2 encodes wrist proximity to the base, and J4–J1 encodes overall arm extension relative to the base. Together, these three connections form a minimal observability basis for the seven-DoF arm posture, providing the operator with intuitive directional feedback about which part of the arm is under strain without requiring them to interpret overlapping or competing force signals. Configuration C also represents the virtual springs present in Kinova Gen3 Arm, Hence C would be the best approach to add on physical figurine.

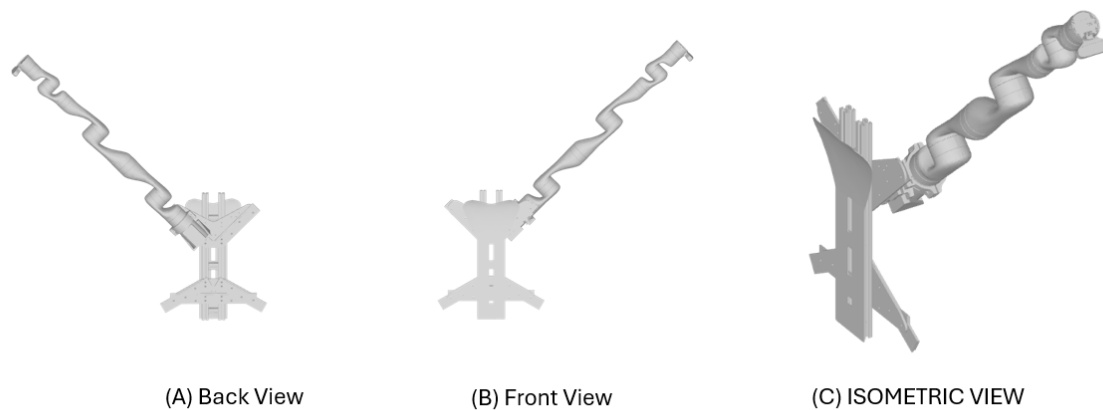


Figure 4: CAD model of the miniature master mannequin arm assembly shown from three perspectives: (A) back view, (B) front view, and (C) isometric view. The model comprises a scaled-down kinematic replica of the Kinova Gen3 7-DoF arm mounted on a symmetric cross-frame base, with coiled spring elements positioned along the arm links to provide haptic force feedback to the operator's fingertips.

The arm structure follows the seven-degree-of-freedom kinematic chain of the Kinova Gen3 arm, scaled down to a size operable by human fingers. Coiled spring elements are integrated along the arm links at the joint locations identified as optimal in the Configuration C Spring placement study. These springs provide the physical compliance that reflects the postural state of the arm back to the operator as a tangible haptic force, without requiring any electronics or sensing at the master stage. The isometric view (C) most clearly illustrates the three-dimensional spatial arrangement of the joints and the relative positioning of the spring elements along the kinematic chain, confirming that the design captures the full workspace geometry of the Kinova arm in miniature form.

8. References

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